With the rapid advancements in technology, specifically artificial intelligence (AI), there has been major discussion around sentience and superintelligence. Today, AI has immediate relevance in our day-to-day lives, from Apple's Siri helping users pick up calls and providing weather updates based on voice commands to ChatGPT answering whatever question imaginable immediately. However, while these breakthroughs are an impressive feat of human intelligence, there has been discourse around AI development and its pace of progress. Although there are differing opinions within both the public and the scientific community, there is a growing apprehension among the public where it is believed that the speed of AI intelligence development and the possibility of a human extinctionist result needs to be addressed immediately. Alternatively, the scientific community holds a different outlook. They offer a more holistic perspective where an AI that surpasses human intelligence is not in the near future, but that current concerns should be on minimizing risks in current AI systems.

To discuss the varying perspectives on the momentum of AI advancement, it is necessary to preface with the cycle of AI research and growth. According to computer scientist Davis Professor of Complexity at the Santa Fe Institute Melanie Mitchell's book "Artificial Intelligence: A Guide for Thinking Humans," there is a 5-10 year, two-part cycle of bubbles and crashes in the AI industry. This cycle greatly influences public perception and scientific progress. The first part, also known as AI spring, is when new ideas generate vast amounts of optimism in the scientific research community, government organizations, and venture capitalists, leading to billions of dollars in funding and fanfare. The second part, also known as AI winter, occurs when promised progress doesn't occur or is less impressive than expectations, causing funding to dry up and, therefore, a lull in AI research. Thus, it can be said that the field of AI has been experiencing a fruitful AI spring by the mid-2000s and beyond.

Public perspective

In the last decade or so, accomplishments in AI such as Google Translate, Amazon Alexa, self-driving cars, and YouTube automated subtitles have fueled the growth of the field, with all of the largest technology companies pouring billions of dollars into AI research and development. Additionally, startup companies that are popping up everywhere are essentially built upon the idea of "take X and add AI." Nonetheless, these advancements come with concerns of sentience and superintelligence and, therefore, AI taking over the world. As defined by the University of Oxford Swedish philosopher Nick Bostrom, superintelligence is "an intellect that is much smarter than the best human brains in practically every field, including scientific creativity, general wisdom and social skills," and such definitions have led to public fears. Perspectives within the public include the fear of AI gaining consciousness and its own thoughts, then self-improving itself over and over again until it reaches a level of intelligence that cannot be matched by human minds, leading to an all-powerful, unfeeling AI that will take over the world.

From May to July 2023, the Artificial Intelligence, Morality, and Sentience (AIMS) Survey was conducted by the Sentience Institute to explore the attitudes of 1,099 United States adults of various backgrounds on AI risks and sentience. It was found that 48.9% believed that AI development is too fast and 51.5% had concerns that AI will cause human extinction. Of the sample, over half believe in the possibility of sentient AIs or were not sure (74.1%) and 74.8% believe that AIs will exceed human intelligence.

Media representations have played a significant role in shaping such a viewpoint. Science fiction films such as "The Terminator" and "2001: A Space Odyssey" portray AI systems as evil beings with the intention to do harm. Consequently, a negative connotation has been associated with AI advancement. On TikTok, X (formerly known as Twitter), and other social media platforms, trending media such as an AI robot seeing itself for the first time and a Tesla robot attacking a Tesla engineer has created concern about how close our society is to creating AI with a mind of its own. Although these videos are unsettling and it may seem that these robots have a mind of their own, the scientific consensus around AI sentience is that current AI systems are not conscious, or even close to being sentient. The explanation for the former video is that its reaction was likely pre-programmed to mimic a human response and the latter video depicts an error in engineering, similar to errors in building a faulty printer or such. Of the aforementioned AIMS Survey, 66.9% identified as science fiction fans and 54.0% agreed that reality matches science fiction.

Another factor that fuels the perspective of immediate AI sentience is the speed of AI progress in recent years. American computer scientist, author, and former Google Director of Engineering — now Google's Principal Researcher and AI Visionary — Ray Kurzweil, is also a futurist most well-known for his prediction of the singularity. The singularity is the point in time in which the pace of technological change will be so rapid and deep — when AI exceeds human intelligence — that human life will forever be transformed. Kurzweil's predictions are founded on the idea of "exponential progress" in the fields of science and technology, especially computers. This is where technological growth seems slow and steady, until there is a sudden and unexpected surge ahead, following the exponential curve, and Kurzweil believes that society will enter that part of the exponential curve before 2050. It's undeniable that Kurzweil has been and is at the optimistic forefront of AI development, boasting an 86% accuracy rate in his technological forecasts. However, he doesn't specify how this singularity will affect humanity, contributing to the public perspective of AI being an immediate concern.

One of the possible products of the singularity is sentient AI. Sentience, according to Gibert & Martin (2022), is "the ability to have subjective experience, which includes perceiving and experiencing things." The biggest example of the possibility of AI sentience occurring shortly happened in 2023 when former Google engineer Blake Lemoine claimed that Google chatbot LaMDA was sentient. Conversations from Lemoine's interview with LaMDA involved the chatbot talking about its emotions, soul, and fear of being turned off, igniting uneasiness in the public and taking social media by storm. However, LaMDA is a large language model whose primary task is to recognize and generate human-like text based on the accumulation of data from all corners of the internet — which is exactly what LaMDA did in its conversations with

Lemoine. Thus, Lemoine's claims in LaMDA being sentient is an example of confirmation bias where people will believe what they are looking for and want to believe.

One of the most influential people in the world who is often associated with technology is Tesla and SpaceX founder, Elon Musk. Musk is also one of the most vocal advocates for the safety and regulation of the future of AI, being one of the co-founders of OpenAI along with entrepreneur Sam Altman. If such a prominent and AI-knowledgeable figure is concerned that "self-learning AI systems might turn hostile to the human species," it's only natural for the public to heavily weigh or consider his concerns. But, by investigating more closely into Musk's worries, it can be found that his immediate priorities are that of AI systems being trained to spread disinformation, biased reporting, and financial scams rather than immediate superintelligence. Such risks serve as the most prominent and influential component in AI models becoming hostile, and these are what the scientific community believes needs to be mitigated first.

Scientific perspective

While AI experts share the public concerns of AI leading to the end of humanity, they are much more focused on the rate of growth that the field is experiencing, and the possible risk outcomes that may come with such rapid advancement. A survey conducted by Grace et al. (2024) asked 2,778 AI researchers who had published in top-tier AI venues for their predictions on the rate of AI advancement and the impact of advanced AI. Participants were asked to predict when 39 milestones in AI development would become feasible, from AI models being able to create a fake new song to replicating a machine learning paper, and all but 4 tasks were predicted to have at least a 50% chance of occurring within the next 10 years. The aggregate 2023 forecast for achieving high-level machine intelligence (HLMI) where machines can accomplish everything a human can, but better, was predicted to be 10% by 2027. Continuing down the timeline of AI progress next would be the possibility of an intelligence explosion, similar to that of Kurzweil's singularity, in which the median participant in the survey doesn't expect rapid AI progress from it. While it is inarguable that AI technology is advancing and AI intelligence surpassing human intelligence cannot be ruled out, it is not an imminent problem. It was found that more than half of the researchers "suggested that 'substantial' or 'extreme' concern is warranted about six different AI-related scenarios, including spread of false information, authoritarian population control, and worsened equality."

AI technologies, especially those that generate realistic media, can be misused for misinformation purposes to influence public opinion, control, and trust. It can also be leveraged to monitor and keep surveillance on populations through the use of facial recognition, resulting in a plethora of ethical issues such as invasion of privacy and human rights. The scientific opinion of the Grace et al. survey is that the research focus and development should be directed toward combating such information security risks rather than the speculative fears of AI domination, and this would require understanding the very foundation of AI systems.

There are many subsets within AI, such as machine learning, deep learning, and natural language processing, among others, which all rely on repetitive learning. AI systems use algorithms that are trained on numerous data sets, also known as training sets, over and over again. The algorithms then adapt for accuracy by "learning from" the training sets and recognizing patterns in the data, essentially recoding itself (Bini, 2018). Once these algorithms are finely tuned by the training sets, the AI system can be implemented in real-world applications to make decisions for the task it is being trained on. AI systems are also categorized under two definitions: narrow and general. Narrow AI systems are those that can perform only one narrowly defined task, or a small set of related tasks, and cannot be applied or adapted to other tasks. On the other hand, general/"strong"/human-level AI systems are those that can do everything humans do, and possibly more and even better, such as the ones that are seen in science fiction films. Although there have been recent breakthroughs in deep learning and neural workings, all AI systems to date are examples of "narrow" or "weak" AI, according to Baum (2018). Understanding the workings of AI clarifies that current AI technologies are highly specialized and far from being sentient, as the algorithms are purely dependent on the data that they are trained on for a specific task. Although the awareness of AI being far from gaining consciousness can alleviate public fears about alarming scenarios such as AI taking over the world, the repetitive learning method is extremely prone to biases.

In a The New York Times opinion article, University of Southern California research professor and Senior Research Professor at Microsoft Research Lab Kate Crawford argued that the debate around superintelligence only diverts attention from more pressing and impactful AI issues, especially AI systems that exacerbate inequalities. Echoing this sentiment, biases are a major concern in AI systems among the AI community, such as the biases of Diffusion AI and how it perpetuates stereotypes to an extreme. Biases can arise when algorithms are being trained in an AI system. If the system is trained on datasets that underrepresent certain groups or contain stereotypes, the system will replicate and only recognize such perspectives.This can lead to a Butterfly Effect — the idea that the world is interconnected and that one small change can leave a great influence — where biases in data, programming, and algorithm training can lead to an AI evolution that negatively impacts marginalized groups and leaves a detrimental mark on both society and future AI models (Ferrera, 2024). Such prejudices in AI systems can emerge in real-life applications such as job applications and criminal profiling. Thus, it has been of utmost importance for AI researchers to solve such matters, and to have full control over AI as it grows. This can be addressed by firstly recognizing that AI operates within a wider societal, human context.

Introducing one of AI's greatest issues: the control problem. The control problem is the challenge of ensuring that the AI systems that are being created will align with humanity's values, and this is one of the largest factors in the fear of hostile AI. However, Agar (2016), argues that the control problem can be and will be resolved, due to its nature of being a "predictably soluble problem," such as programming human-friendly values into AI systems. If AI can be taught human values, fears will be mitigated in regard to an existential threat. Yet, present AI programs are still far from superintelligence, or even sentience, as they are all weak AI, as mentioned previously. Many do not believe that AI will fully replicate human empathy (Perry, 2023), as a major theory to achieve natural intelligence is through experiential learning, just like how humans do. Experiential learning is learning through experiences and interactions, in which one generates their own thoughts and outcomes. This not only takes time, but empathy also requires cognitive empathy (understanding others' emotional states), emotional empathy (experience sharing), and motivational empathy (empathic care). This adds another extremely challenging characteristic in attempting to achieve general intelligence, as the complexities of human cognition and emotions are seemingly impossible to replicate in machines. As of now, no AI has been created yet that is naturally intelligent. Therefore, it is not yet time to worry about superintelligence as existing models are far from achieving AI that can create its own logic and have its own emotions.

At Dartmouth College during the summer of 1856, during the conception of AI, the goal of building a naturally intelligent machine within a decade was formed. There are two sides to it: the scientific side of trying to embed the mechanisms of "natural" (biological) intelligence in computers, and the practical side of creating computer programs that outperform humans in specific tasks without worrying about whether these programs mimic human thought processes or not. Today, over half a century later, the industry is still striving to achieve that goal and, as the field has evolved, the definition of the AI field has changed with it. Now, AI is referred to as "a branch of computer science that studies the properties of intelligence by synthesizing intelligence (Mitchell, 2019)." While AI has made remarkable strides, the cycle of AI research, marked by periods of optimism and disappointment, suggests that there are still a number of cycles that the industry has yet to endure in order to make progress toward the original goal.

References

Agar, N. (2016). Don’t worry about superintelligence. Journal of Ethics and Emerging Technologies, 26(1), 73-82

Baum, S. D. (2018). Superintelligence skepticism as a political tool. Information, 9(9), 209

Bini, S. A. (2018). Artificial intelligence, machine learning, deep learning, and cognitive computing: what do these terms mean and how will they impact health care?. The Journal of arthroplasty, 33(8), 2358-2361.

Bostrom, N. (1998). How long before superintelligence. International Journal of Futures Studies, 2(2003), 12-17

Ferrara, E. (2024). The butterfly effect in artificial intelligence systems: implications for AI bias and fairness. Machine Learning with Applications, 15, 100525.

Grace, K., Stewart, H., Sandkühler, J. F., Thomas, S., Weinstein-Raun, B., & Brauner, J. (2024). Thousands of AI Authors on the Future of AI. arXiv preprint arXiv:2401.02843.

Mitchell, M. (2019). *Artificial intelligence: A guide for thinking humans*. Penguin UK.

Pauketat, J. V., Bullock, J., & Anthis, J. R. (2023, September 8). Public Opinion on AI Safety: AIMS 2023 Supplement.

Perry, A. (2023). AI will never convey the essence of human empathy. Nature Human Behaviour, 7(11), 1808-1809.